ME 5309 – Mid-Semester Examination: ANSWER SHEET

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Questions attempted: Q1, Q2, Q3

Question 1

(a) (i) $f_x = f/p = 16/5 =$ **3.2 px**.

(ii) Width of sensor = 1280 × 5 = 6400. FOV = $N \cdot Z/f_x = 1280 \times 400 / 3.2 =$ **160000 mm**.

One pixel = FOV/N = 160000/1280 = **125 mm per pixel**. (looks large but the formula is right)

(b) LSB = 5/4096 = 1.2207 mV.

ΔF = LSB / S = 1.2207 mV / 10 mV/N = **0.0122 N**.

(c) $F = e^{AT}$ and $G = \int_0^T e^{A\tau} B \, d\tau$.

$F = e^{-0.2} =$ **0.8187**.

$G = \int_0^{0.1} e^{-2\tau} \, d\tau = 1 - e^{-0.2} =$ **0.1813**.

Question 2

(a) States $x_1 = \theta$, $x_2 = \dot{\theta}$. $\dot{x}_1 = x_2$; $\dot{x}_2 = (1/J)[\tau - b\, x_2 - mg\ell \sin x_1]$.

Equilibria ($\tau = \tau_0$, $\dot{x} = 0$): $x_2 = 0$ and $mg\ell \sin \theta_e = \tau_0$. Sence $|\sin \theta| \le 1$, an equilibrium exists iff $|\tau_0| \le mg\ell = 2.4525$ N·m.

(b) Linearising: A = [0, 1; $\partial f_2/\partial x_1$, $\partial f_2/\partial x_2$].

$\partial f_2/\partial x_1 = -(mg\ell/J) \sin \theta_e = -122.625 \times 0.8660 = -106.2$. $\partial f_2/\partial x_2 = -b/J = -2.5$.

A = [0, 1; −106.2, −2.5], B = [0; 1/J] = **[0; 50]**.

(c) At rest $\dot{\theta} = \ddot{\theta} = 0$, so the PD torque must balance gravity: $K_p(\theta_d - \theta_{ss}) = mg\ell \sin \theta_{ss}$.

If $\theta_{ss} = \theta_d = 90°$ the left side is 0 but the right side is $mg\ell \sin 90° = 2.4525 \neq 0$: contradiction. So the joint cannot rest at θ_d; the gravity torqe needs a non-zero error to produce the holding torqe.

Solve $50(\pi/2 - \theta_{ss}) = 2.4525 \sin \theta_{ss}$ numerically: $\theta_{ss} \approx 1.5218$ rad = 87.19°. Error ≈ 90° − 87.19° = **2.81°** (joint sags below θ_d).

(d) With compensation, at rest: $K_p(\theta_d - \theta_{ss}) + mg\ell \sin \theta_{ss} = mg\ell \sin \theta_{ss} \Rightarrow \theta_{ss} = \theta_d$: **steady-state error = 0**.

Closed loop: $J\ddot{\theta} + (b + K_d)\dot{\theta} + K_p(\theta - \theta_d) = 0$; gravity cancels. Without compensaton the linearised stiffness is $K_p + mg\ell \cos \theta_d$, and at $\theta_d = 90°$ $\cos \theta_d = 0$, so the stiffness is K_p either way. **Poles unchanged** (specific to 90°; at other angles they would shift).

Question 3

(a) (1) Finite differencing introduces an extra pole wich makes the plant unstable.

(2) It cannot be implemented digitally because the derivative of a sampled signal is not defined.

(b) $H(s) = \omega_c/(s + \omega_c)$, $\omega_c = 2\pi(20) = 125.66$ rad/s. Tustin: $s = (2/T)(z - 1)/(z + 1)$, T = 1 ms. With $a = \omega_c T/2 = 0.06283$: $H(z) = a(1 + z^{-1}) / [(1 + a) - (1 - a)z^{-1}]$.

$y_k = [a/(1+a)](u_k + u_{k-1}) + [(1-a)/(1+a)]\, y_{k-1} =$ **$0.05912\,(u_k + u_{k-1}) + 0.88177\, y_{k-1}$**.

DC gain: $z = 1$ gives $H(1) = 2(0.05912)/(1 - 0.88177) = 0.11824/0.11823 = 1.000$ ✓.

(c) $A - LC = [-\ell_1, 1; -\ell_2, -2]$. $\det(sI - A + LC) = (s + \ell_1)(s + 2) + \ell_2 = s^2 + (2 + \ell_1)s + (2\ell_1 + \ell_2)$.

Desired: $(s + 10)^2 = s^2 + 20s + 100$. Match: $2 + \ell_1 = 20 \Rightarrow \ell_1 = 18$; $2\ell_1 + \ell_2 = 100 \Rightarrow \ell_2 = 64$. **$L = [18, 64]^T$**.

(d) Advantage of the obsever: it uses the plant model and the applied input u, so its velocity estimate has (almost) no phase lag; the 20 Hz filter on differenced data always delays by about $1/(2\pi f_c) \approx 8$ ms. It can also reconstruct states that cannot be obtained by differencing (e.g. disturbance, when augmented).

Fliter is safer when the model is uncertain or changes (unknown payload, friction, contact, acuator saturation): the observer then gives a biased or poorly-behaved estimate while the model-free filter still follows the measured data.